Exam AWS Certified Developer - Associate DVA-C02 All Questions

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EXAM AWS CERTIFIED DEVELOPER - ASSOCIATE DVA-C02 TOPIC 1 QUESTIO...

Exam question from Amazon's AWS Certified Developer - Associate DVA-C02
Question #: 260
Topic #: 1
[All AWS Certified Developer - Associate DVA-C02 Questions]

A developer is writing an application that will retrieve sensitive data from a third-party system. The application will format the data into a PDF file. The PDF file could be more than 1 MB. The application will encrypt the data to disk by using AWS Key Management Service (AWS KMS). The application will decrypt the file when a user requests to download it. The retrieval and formatting portions of the application are complete.

The developer needs to use the GenerateDataKey API to encrypt the PDF file so that the PDF file can be decrypted later. The developer needs to use an AWS KMS symmetric customer managed key for encryption.

Which solutions will meet these requirements?

A. Write the encrypted key from the GenerateDataKey API to disk for later use. Use the plaintext key from the GenerateDataKey API and a symmetric encryption algorithm to encrypt the file.

B. Write the plain text key from the GenerateDataKey API to disk for later use. Use the encrypted key from the GenerateDataKey API and a symmetric encryption algorithm to encrypt the file.

C. Write the encrypted key from the GenerateDataKey API to disk for later use. Use the plaintext key from the GenerateDataKey API to encrypt the file by using the KMS Encrypt API.

D. Write the plain text key from the GenerateDataKey API to disk for later use. Use the encrypted key from the GenerateDataKey API to encrypt the file by using the KMS Encrypt API.

Show Suggested Answer

by CrescentShared at *Feb. 21, 2024, 3:17 a.m.*

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Comments

CrescentShared Highly Voted 2 years, 1 month ago

Selected Answer: A

Using the KMS Encrypt API to encrypt large amounts of data, such as a PDF file that could be more than 1 MB, is not efficient and can be costly. AWS KMS is designed for encrypting small amounts of data, such as encryption keys or short strings. For larger data, it's recommended to use a client-side encryption library with a data key generated by KMS.

upvoted 7 times

fsk8 Most Recent 7 months, 2 weeks ago

Selected Answer: A

A - This is the standard and secure way to use GenerateDataKey. You use the plaintext data key to encrypt your large data locally, and store the encrypted data key alongside the encrypted data.
B - You should never write the plaintext key to disk. The whole point of KMS is to protect the plaintext key.
C - You use the plaintext data key to encrypt the file locally, not by calling KMS Encrypt for the entire file as KMS Encrypt limited to 4KB - https://docs.aws.amazon.com/kms/latest/APIReference/API_Encrypt.html
D - Similar to B, you should never store the plaintext key on disk, and similar to C, you encrypt the file locally with the plaintext data key, not by calling KMS Encrypt on the file itself.

upvoted 1 times

0bdf3af 1 year ago

Selected Answer: C

C. We use KMS Encrypt API and this method is called envelope encyrption. KMS will generate plaintext key which we have to store on the disk. We use it to encrypt file by calling KMS API

upvoted 1 times

preachr 1 year, 6 months ago

Selected Answer: A

To encrypt data outside of AWS KMS:

1) Use the GenerateDataKey operation to get a data key.

2) Use the plaintext data key (in the Plaintext field of the response) to

encrypt your data outside of AWS KMS. Then erase the plaintext data key from memory.

3) Store the encrypted data key (in the CiphertextBlob field of the response) with the encrypted data.

upvoted 2 times

wh1t4k3r 1 year, 7 months ago

Selected Answer: C

Where is the KMS key element on A?

upvoted 1 times

jyrajan69 1 year, 8 months ago

The question clearly says using KMS so why would you even consider A and B

upvoted 2 times

65703c1 1 year, 10 months ago

Selected Answer: A

A is the correct answer.

upvoted 2 times

DeaconStJohn 2 years ago

Selected Answer: C

Going with my gut.

upvoted 1 times

SerialiDr 2 years ago

Selected Answer: A

Option A is the most appropriate method for encrypting a PDF file using AWS KMS, where the plaintext key is used for encryption operations, and the encrypted key (not the plaintext key) is stored or managed externally for later decryption use.

upvoted 2 times

Abdullah22 2 years ago

going with C

upvoted 1 times

KarBiswa 2 years, 1 month ago

Selected Answer: C

https://docs.aws.amazon.com/kms/latest/APIReference/API_GenerateDataKey.html

upvoted 3 times

KarBiswa 2 years, 1 month ago

Sorry changing it to A as C is not about saying using plain text data as customized. Though the algorithm word is confusing

upvoted 1 times

KarBiswa 2 years ago

C is the option final decision

upvoted 1 times

ANDRES715 2 years, 1 month ago

Selected Answer: D

Según la documentación de AWS, cuando se utiliza la API GenerateDataKey, se obtiene una clave de texto sin formato y una clave cifrada. La clave de texto sin formato se puede escribir en el disco para su uso posterior, mientras que la clave cifrada se utiliza para cifrar los datos. En este caso, el desarrollador debe escribir la clave de texto sin formato en el disco para su uso posterior y utilizar la clave cifrada para cifrar el archivo PDF mediante la API de cifrado KMS.

upvoted 1 times